

CSE 208: Data Structures and Algorithms II

Online on Flow & Matching

Department of Computer Science and Engineering

Term: July 2025

Time: 45 Minutes

The city of Dhaka is suffering from severe traffic congestion. The city officials have modeled the road network as a directed graph where intersections are nodes and roads are directed edges with a specific vehicle capacity (vehicles/hour).

Currently, the traffic flow from the Uttara to Motijheel is strictly limited by congested roads. The officials have a budget to construct exactly **one** new flyover. They have a list of P proposed flyover projects. Each proposal connects two existing intersections with a new high-capacity road.

Task Requirements

Your task is to identify which of the proposed flyovers would strictly increase the maximum traffic flow if constructed. If multiple projects work, list all of them.

Input Format

- The first line of input will have two space-separated non-negative integers N and M , the number of intersections and existing roads.
- In the next M lines, there will be three space-separated integers, u , v , c denoting an existing directed road from u to v with capacity c .
- The next line will contain an integer P , the number of proposed flyover projects.
- In the next P lines, there will be three space-separated integers u , v , c representing a potential new flyover from u to v with capacity c .
- Assume the Source is always node 1 and the Sink is always node N .

Output Format

- Print a space-separated list of the **indices** (corresponding to the order in the input) of the proposed flyovers that, if built, would strictly increase the total max flow.
- If no proposal improves the flow, print **None**.

Sample I/O

Input	Output
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4 5
1 2 5
1 3 5
2 3 5
2 4 5
3 4 5
2
2 3 100
1 4 100

2